Computer Networking

Practical no. 01

1] Ping

Definition ➖

The command `ping google.com` is used to test the connectivity between your computer and Google's servers by sending ICMP (Internet Control Message Protocol) Echo Request packets and waiting for Echo Reply packets.

****Explanation**
When you run `ping google.com`, it checks if Google's servers are reachable over the internet and measures the round-trip time (latency) of the packets. This helps diagnose network issues like packet loss or high latency.

Theory:
Ping is a diagnostic tool based on the ICMP protocol, used in computer networking to verify the status of a network connection and troubleshoot connectivity problems by sending packets and evaluating responses.

Command :- pin google.com

```
student@sjchslab4:~$ ping google.com
PING google.com (142.250.183.78) 56(84) bytes of data.
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=1 ttl=120 time=3.67 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=2 ttl=120 time=3.66 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=3 ttl=120 time=3.65 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=4 ttl=120 time=3.93 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=5 ttl=120 time=3.73 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=6 ttl=120 time=3.69 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=7 ttl=120 time=3.78 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=8 ttl=120 time=3.65 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=9 ttl=120 time=3.48 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=10 ttl=120 time=3.62 ms

64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=12 ttl=120 time=3.73 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=13 ttl=120 time=3.65 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=14 ttl=120 time=4.38 ms
64 bytes from bom12s12-in-f14.1e100.net (142.250.183.78): icmp_seq=15 ttl=120 time=4.50 ms
^Z
[1]+  Stopped                 ping google.com
```

2] net stat

Definition:
The command `netstat -an` is used to display all active network connections and listening ports on your computer, showing the addresses and port numbers in numerical form.

Explanation:
When you run `netstat -an`, it provides a list of all open ports and the current state of network connections, including TCP and UDP connections, and their associated local and remote addresses, without resolving hostnames.

Theory:
`netstat` (network statistics) is a command-line tool for network troubleshooting and performance monitoring. The `-an` flags make it show all connections and listening ports in numeric format, which helps in understanding the network configuration and diagnosing issues like unexpected open ports or unauthorized connections.

Command :- `netstat -tunap`

```
student@sjchslab4:~$ netstat -tunap
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:25672           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:5900            0.0.0.0:*               LISTEN      10732/vino-server
tcp        0      0 10.0.4.253:44791        0.0.0.0:*               LISTEN      10729/rygel
tcp        0      0 127.0.0.1:41169         0.0.0.0:*               LISTEN      10729/rygel
tcp        0      0 0.0.0.0:445             0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:139             0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:4330            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -
tcp        0      0 192.168.122.1:42749     0.0.0.0:*               LISTEN      10729/rygel
tcp        0      0 0.0.0.0:44322           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:44323           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:44321           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:11211         0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:27017         0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 192.168.122.1:53        0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:49535         127.0.0.1:4369          TIME_WAIT   -
tcp        0      0 127.0.0.1:46065         127.0.0.1:4369          ESTABLISHED -
tcp        0      0 10.0.4.253:55440        10.0.4.207:5900         ESTABLISHED 10822/remmina
tcp        0      0 10.0.4.253:44791        10.0.10.7:65065         ESTABLISHED 10729/rygel
tcp6       0      0 :::5900                 :::*                    LISTEN      10732/vino-server
tcp6       0      0 :::5672                 :::*                    LISTEN      -
tcp6       0      0 :::445                  :::*                    LISTEN      -
tcp6       0      0 :::4369                 :::*                    LISTEN      -
tcp6       0      0 :::139                  :::*                    LISTEN      -
tcp6       0      0 :::4330                 :::*                    LISTEN      -
tcp6       0      0 :::22                   :::*                    LISTEN      -
tcp6       0      0 :::80                   :::*                    LISTEN      -
tcp6       0      0 :::44322                :::*                    LISTEN      -
tcp6       0      0 :::44323                :::*                    LISTEN      -
tcp6       0      0 :::44321                :::*                    LISTEN      -
tcp6       0      0 :::3306                 :::*                    LISTEN      -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
tcp6       0      0 127.0.0.1:4369          127.0.0.1:46065         ESTABLISHED -
tcp6       0      0 10.0.4.253:4369         10.0.4.253:54527        TIME_WAIT   -
udp        0      0 127.0.0.1:37484         0.0.0.0:*                           10729/rygel
udp        0      0 239.255.255.250:1900    0.0.0.0:*                           10729/rygel
udp        0      0 192.168.122.1:1900      0.0.0.0:*                           10729/rygel
udp        0      0 239.255.255.250:1900    0.0.0.0:*                           10729/rygel
udp        0      0 10.0.4.253:1900         0.0.0.0:*                           10729/rygel
udp        0      0 239.255.255.250:1900    0.0.0.0:*                           10729/rygel
udp        0      0 127.0.0.1:1900          0.0.0.0:*                           10729/rygel
udp        0      0 192.168.122.1:51518     0.0.0.0:*                           10729/rygel
udp        0      0 10.0.4.253:56701        0.0.0.0:*                           10729/rygel
udp        0      0 192.168.122.1:53        0.0.0.0:*                           -
udp        0      0 127.0.0.53:53           0.0.0.0:*                           -
udp        0      0 0.0.0.0:67              0.0.0.0:*                           -
udp        0      0 10.0.4.253:68           10.0.1.20:67            ESTABLISHED -
udp        0      0 10.0.4.253:123          0.0.0.0:*                           -
udp        0      0 127.0.0.1:123           0.0.0.0:*                           -
udp        0      0 0.0.0.0:123             0.0.0.0:*                           -
udp        0      0 192.168.122.255:137     0.0.0.0:*                           -
udp        0      0 192.168.122.1:137       0.0.0.0:*                           -
udp        0      0 10.0.31.255:137         0.0.0.0:*                           -
udp        0      0 10.0.4.253:137          0.0.0.0:*                           -
udp        0      0 0.0.0.0:137             0.0.0.0:*                           -
udp        0      0 192.168.122.255:138     0.0.0.0:*                           -
udp        0      0 192.168.122.1:138       0.0.0.0:*                           -
udp        0      0 10.0.31.255:138         0.0.0.0:*                           -
udp        0      0 10.0.4.253:138          0.0.0.0:*                           -
udp        0      0 0.0.0.0:138             0.0.0.0:*                           -
udp6       0      0 ::1:123                 :::*                                -
udp6       0      0 :::123                  :::*                                -
```

3] arp

Definition:

The command `arp -a` displays the ARP (Address Resolution Protocol) table, showing the mapping of IP addresses to MAC (Media Access Control) addresses on your local network.

Explanation:
When you run `arp -a`, it lists the IP-to-MAC address mappings that your computer knows about, which are used for local network communication. This helps in identifying the physical addresses of devices connected to the same network.

Theory:
ARP is used to map a network layer (IP address) to a data link layer (MAC address). The `arp -a` command helps network administrators troubleshoot and verify the physical address (MAC) of devices associated with specific IP addresses on the local network.

command :- arp -a same for this

```
student@sjchslab4:~$ arp -a
? (10.0.13.4) at 50:91:e3:04:a7:ad [ether] on enp1s0
? (10.0.4.207) at 50:eb:f6:ca:5f:fd [ether] on enp1s0
? (10.0.7.245) at 9c:93:4e:9b:84:0e [ether] on enp1s0
? (10.0.12.120) at e0:be:03:8e:8f:2d [ether] on enp1s0
? (10.0.13.254) at d0:ad:08:c8:80:07 [ether] on enp1s0
? (10.0.3.31) at 8c:89:a5:f6:7f:17 [ether] on enp1s0
? (10.0.10.57) at 10:78:d2:e9:1c:87 [ether] on enp1s0
? (10.0.3.4) at f4:93:9f:f6:89:0b [ether] on enp1s0
? (10.0.7.163) at 00:19:a9:bc:03:d2 [ether] on enp1s0
? (10.0.0.63) at b0:39:56:80:24:b7 [ether] on enp1s0
? (10.0.3.175) at 1c:69:7a:f2:5d:6b [ether] on enp1s0
? (10.0.0.60) at a0:04:60:06:91:5a [ether] on enp1s0
? (10.0.6.189) at 1c:69:7a:4b:23:11 [ether] on enp1s0
_gateway (10.0.1.21) at 18:c2:41:32:11:09 [ether] on enp1s0
? (10.0.1.45) at e8:fc:af:e6:5f:c2 [ether] on enp1s0
? (10.0.10.7) at 00:23:24:f1:d9:8a [ether] on enp1s0
? (10.0.5.124) at d8:43:ae:82:52:61 [ether] on enp1s0
? (10.0.12.200) at 00:a1:59:26:66:9e [ether] on enp1s0
? (10.0.4.88) at 2c:59:e5:74:91:df [ether] on enp1s0
? (10.0.5.222) at e0:73:e7:2d:5d:77 [ether] on enp1s0
? (10.0.18.221) at 08:8f:c3:73:31:fc [ether] on enp1s0
? (10.0.13.8) at 40:ae:30:df:6d:85 [ether] on enp1s0
? (10.0.7.248) at 50:57:9c:4e:35:27 [ether] on enp1s0
? (10.0.0.73) at b0:39:56:80:25:27 [ether] on enp1s0
? (10.0.10.51) at 10:78:d2:e2:f7:49 [ether] on enp1s0
? (10.0.7.35) at 7c:64:6c:a4:ce:46 [ether] on enp1s0
? (10.0.10.71) at 74:c1:7e:66:b8:a9 [ether] on enp1s0
? (10.0.3.233) at 1c:69:7a:4b:29:e1 [ether] on enp1s0
? (10.0.4.246) at f4:93:9f:f5:a2:f5 [ether] on enp1s0
? (10.0.10.69) at 10:78:d2:e6:cf:f4 [ether] on enp1s0
? (10.0.17.73) at 00:e0:4c:68:22:f4 [ether] on enp1s0
? (10.0.1.71) at 7c:64:6c:c4:7b:73 [ether] on enp1s0
? (10.0.17.190) at e4:a8:df:e3:cf:7c [ether] on enp1s0
? (10.0.6.10) at d8:5e:d3:45:cc:05 [ether] on enp1s0
? (10.0.12.98) at e0:be:03:8e:ac:fa [ether] on enp1s0
? (10.0.1.250) at 38:9d:92:e1:bd:4b [ether] on enp1s0
? (10.0.10.61) at 10:78:d2:e2:f7:2a [ether] on enp1s0
```

4] **ipconfig (Windows) / ifconfig (Linux)**

Definition:
The command `ifconfig` is used to display or configure the network interfaces on a Linux or Unix-based system, showing details like IP addresses, netmask, broadcast address, and interface status.

Explanation:

When you run `ifconfig`, it lists all network interfaces on your system (e.g., Ethernet, Wi-Fi), along with their current settings such as IP address, MAC address, and data transfer statistics. It can also be used to enable or disable interfaces.

Theory:
`ifconfig` (interface configuration) is a system administration tool that allows users to view and manage network interface configurations. It's useful for troubleshooting network connectivity, viewing network stats, or assigning new IP addresses or subnet masks to interfaces. On newer Linux systems, `ip` command has replaced `ifconfig` for more advanced features.

command :- ifconfig

```
student@sjchslab4:~$ ifconfig
enp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.4.253  netmask 255.255.224.0  broadcast 10.0.31.255
        inet6 fe80::372a:ecd5:ce48:3467  prefixlen 64  scopeid 0x20<link>
        inet6 fe80::c689:302d:7167:8852  prefixlen 64  scopeid 0x20<link>
        inet6 fe80::2372:e11c:8726:af9a  prefixlen 64  scopeid 0x20<link>
        ether f4:93:9f:f3:ec:cc  txqueuelen 1000  (Ethernet)
        RX packets 139452  bytes 113923132 (113.9 MB)
        RX errors 0  dropped 732  overruns 0  frame 0
        TX packets 26518  bytes 2138342 (2.1 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 877  bytes 121293 (121.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 877  bytes 121293 (121.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.122.1  netmask 255.255.255.0  broadcast 192.168.122.255
        ether 52:54:00:7f:09:59  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

5] **hostname**

Definition:

The command `hostname` is used to display the name of the computer or device on a network.

Explanation:
When you run `hostname`, it shows the system's assigned network name, which helps identify the device within a network and is often used in system configuration or troubleshooting.

Theory:
A hostname is a label assigned to a device connected to a network, used for identification and communication. The `hostname` command helps administrators verify or change this name, ensuring proper network organization and device recognition.

command :- Hostname

```
student@sjchslab4:~$ hostname
sjchslab4
```

6] **nslookup**

****Definition:****
The command `nslookup google.com` is used to query the Domain Name System (DNS) to retrieve the IP address associated with the domain name "google.com."

****Explanation:****
When you run `nslookup google.com`, it sends a request to a DNS server to resolve the domain name "google.com" into its corresponding IP address, allowing your computer to connect to the site.

****Theory:****
`nslookup` (Name Server Lookup) is a network utility used to query DNS servers and obtain information about domain names, such as IP addresses, mail servers, and DNS records. It helps troubleshoot DNS-related issues and verify domain-to-IP resolutions.

command :- nslookup google.com

```
student@sjchslab4:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.42.238
Name:   google.com
Address: 2404:6800:4009:80e::200e
```

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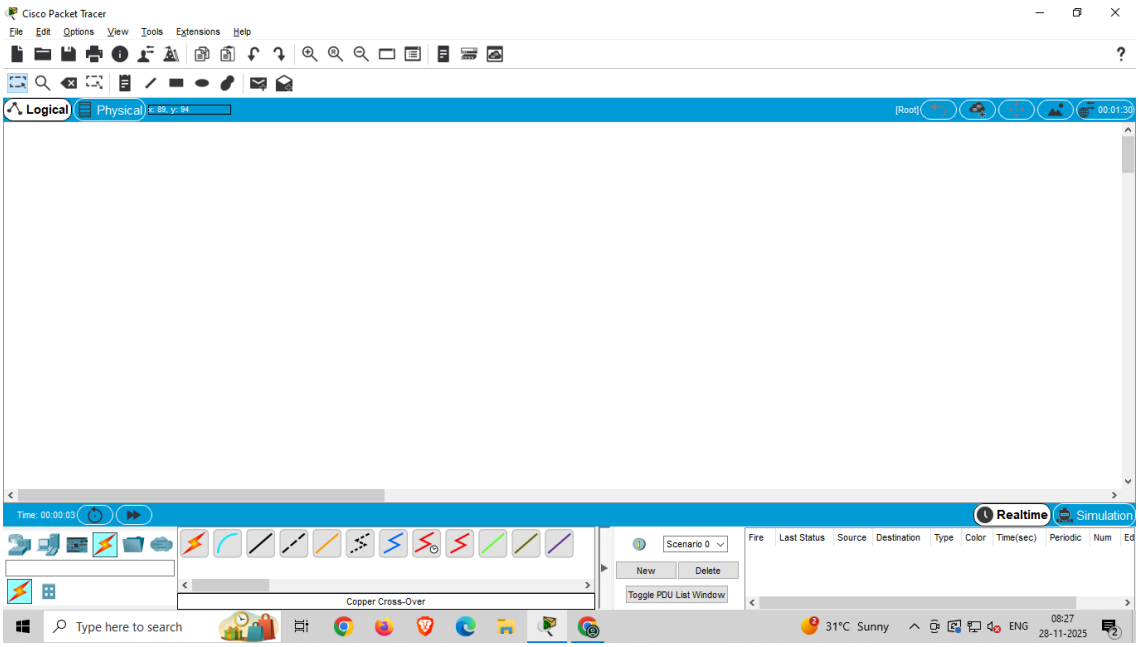
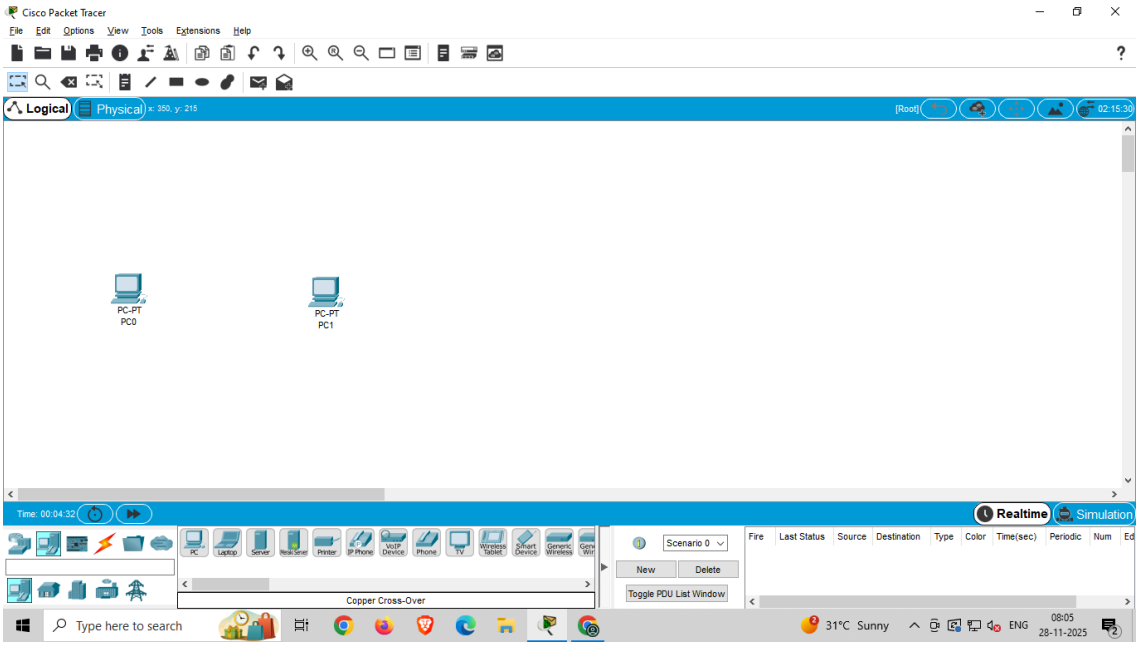
PRACTICAL NO.2

AIM:- Using a packet tracer, create a basic network of two computers using an appropriate network using a static address network.

Step 1:- Cisco Packet Tracer

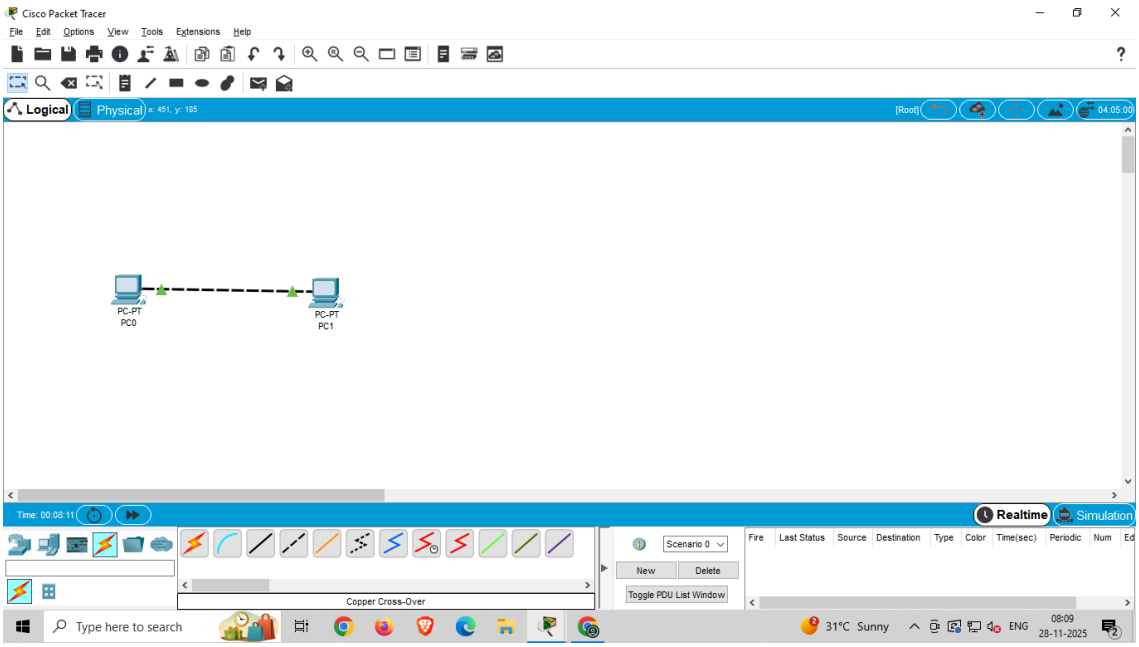


Step 2:- Click on “End Device”
Step 3:- then drag two pc on screen

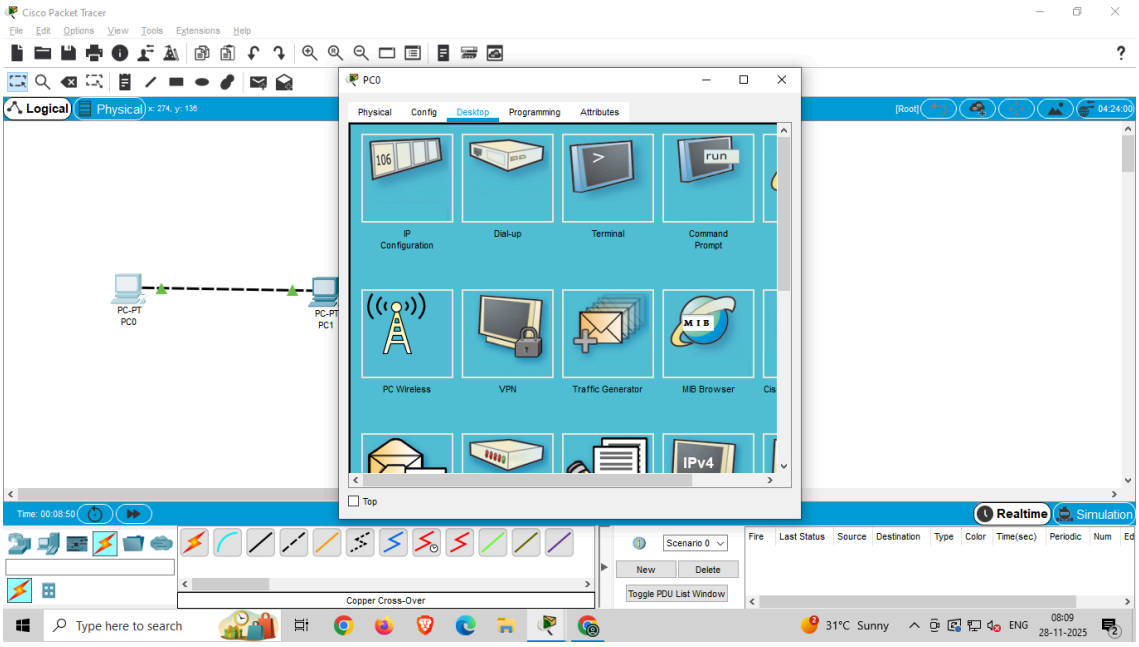


Step4:- click on “Connections”

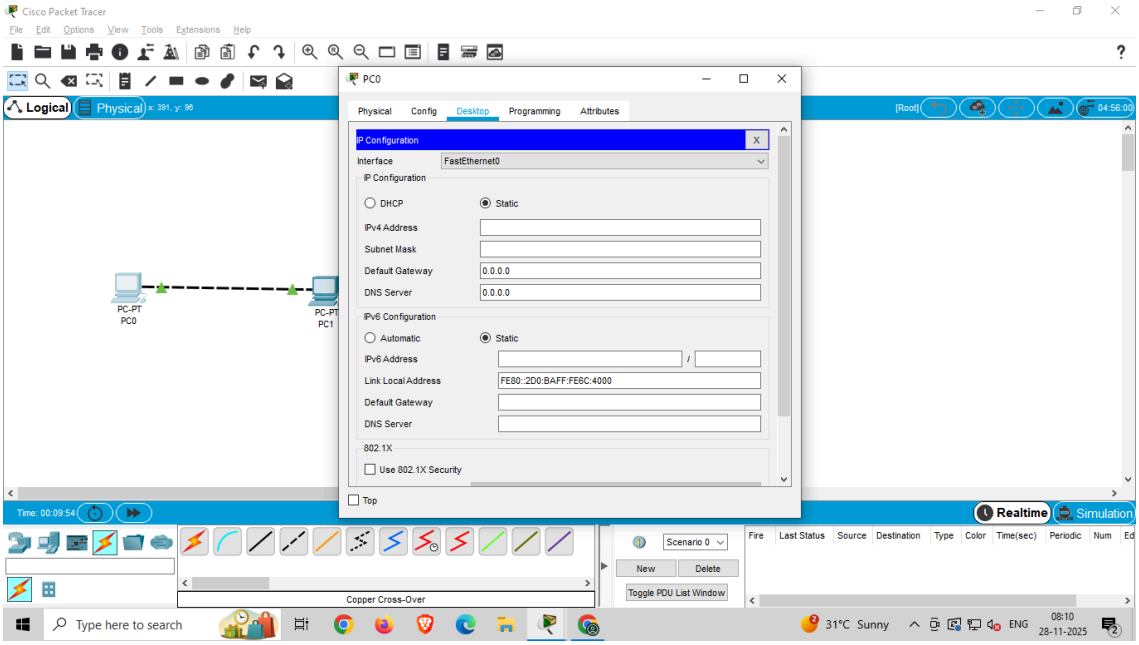
Step5:- after that click on “copper cross-over”, the connect that with both the pc’s



Step6:- Then click on first PC

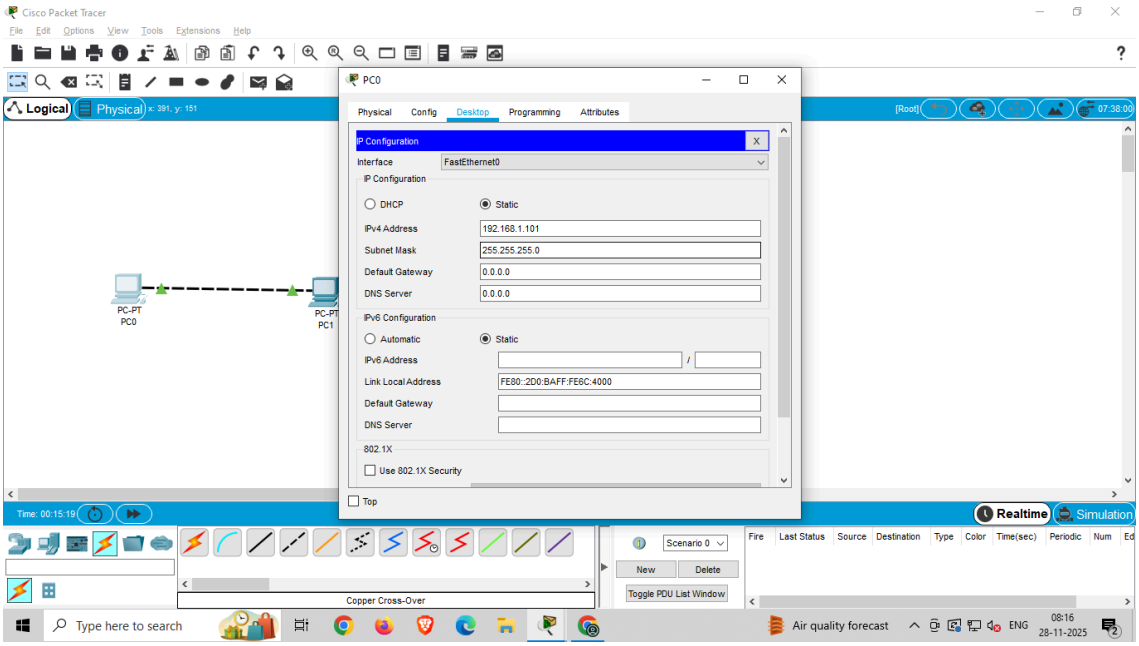


Then go on “deskktop”, then tap on pc configuration



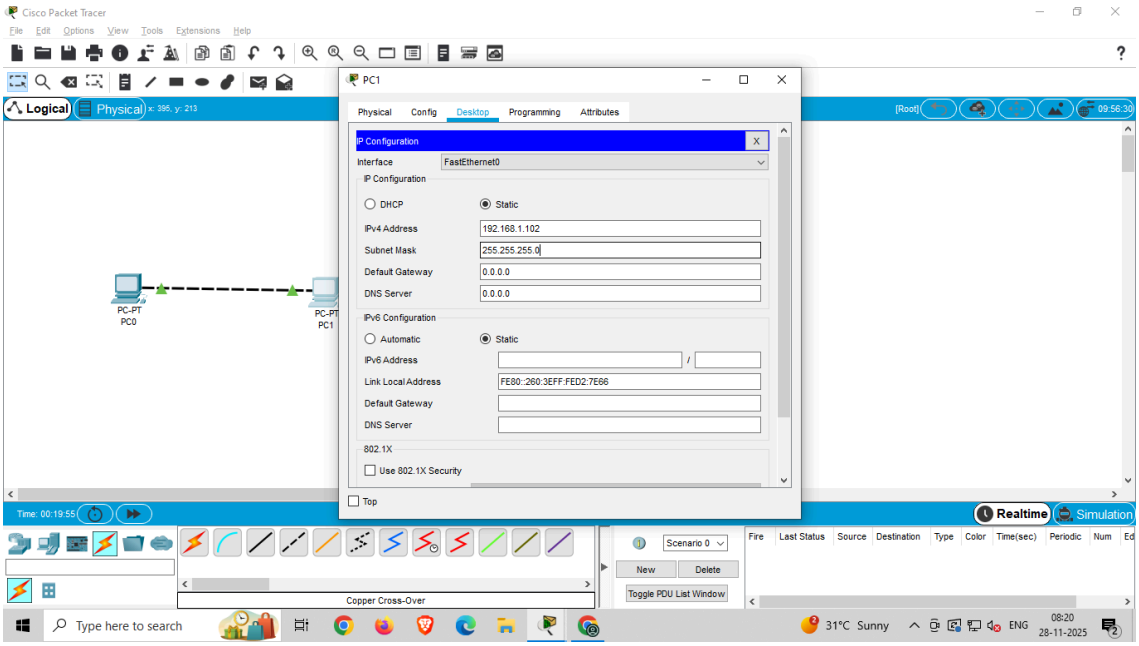
Then click on IPv4 address, add = 192.168.1.10

Then click on Subnet Mask, the value will add automatically

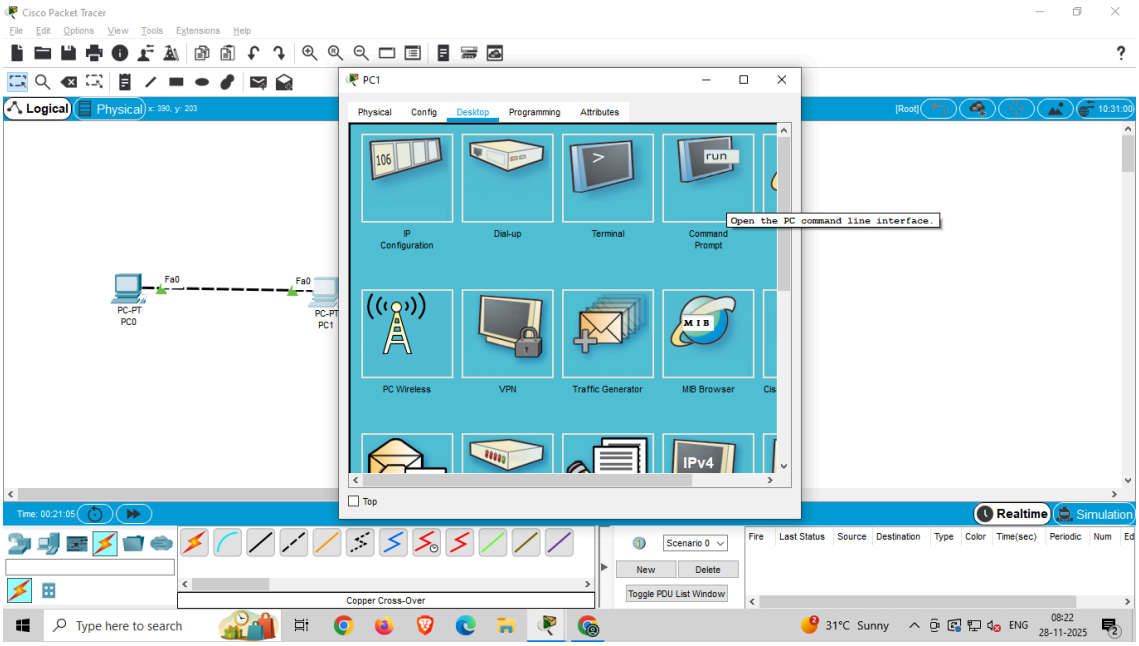


Step 7:- now same do for another pc

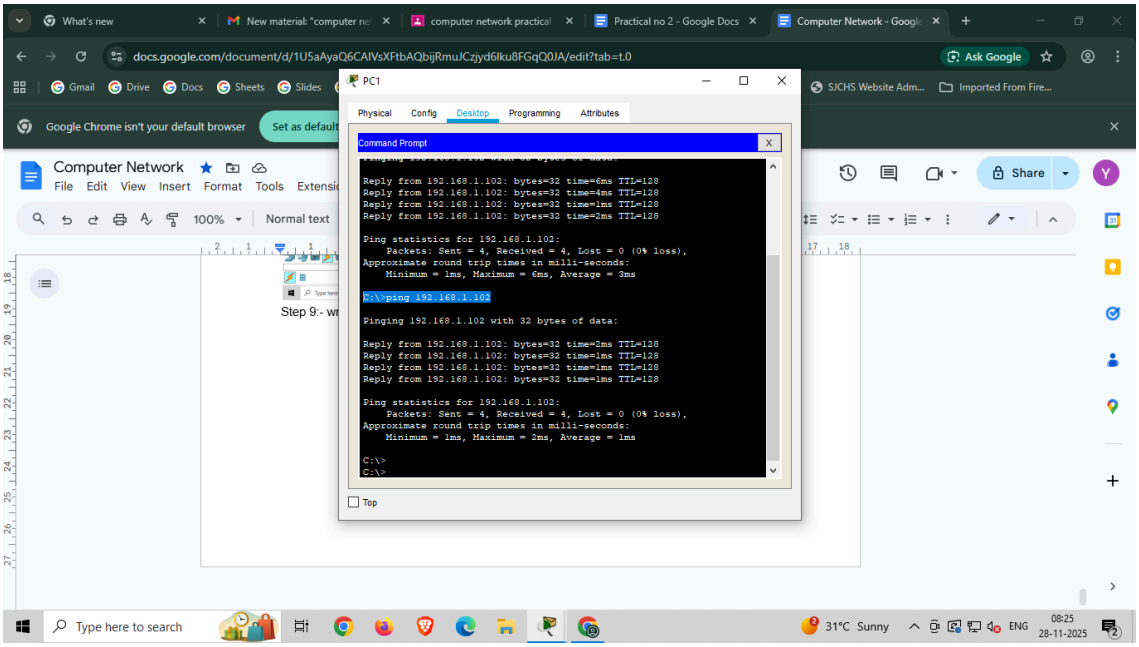
Add IPv4 Address :- 192.168.1.102



Step 8:- Now tap now Second pc the goo on desktop then tap on command prompt



Step 9:- write this prompt " C:\>ping 192.168.1.102 "



Your both the pc's are connected.

Practical 3:-

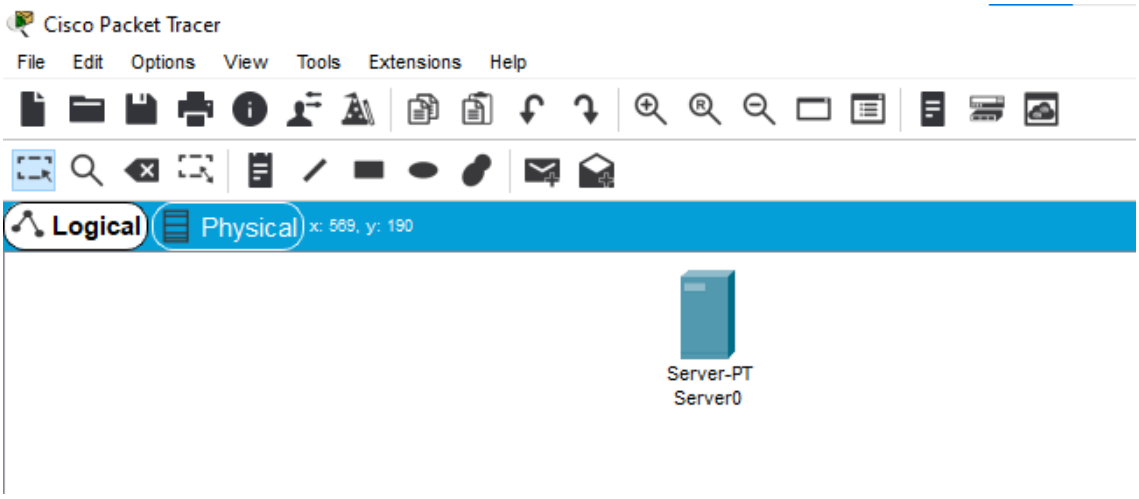
AIM

To configure a network with **one server** and **two client computers** using a **switch**, assign **static IP addresses**, enable **server services (HTTP/FTP)**, and test connectivity between all devices.:

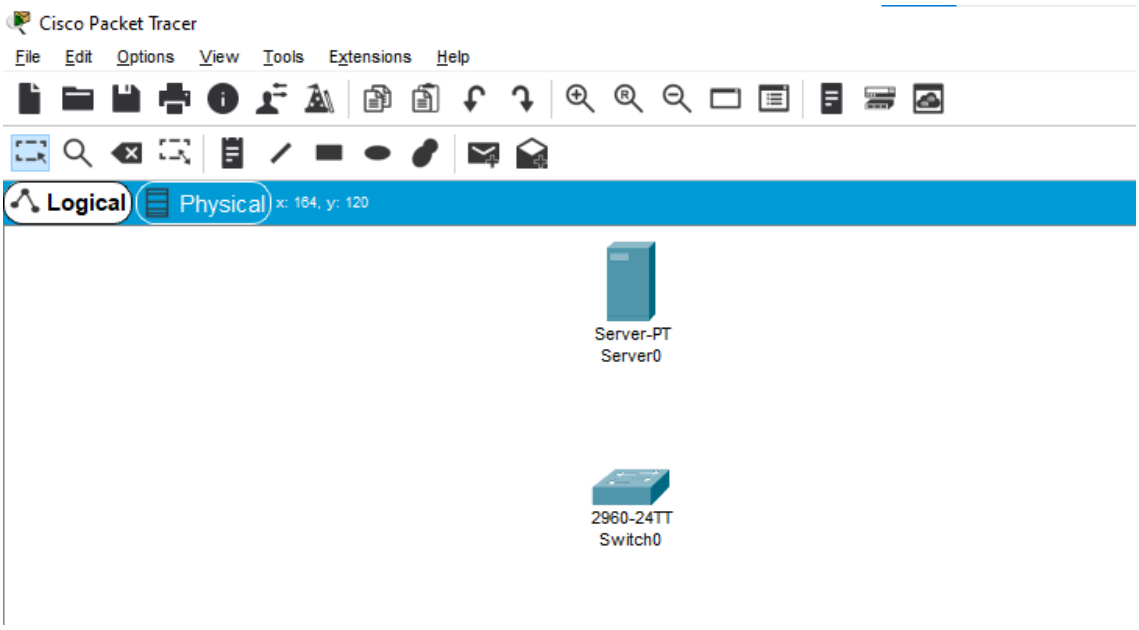
APPARATUS / SOFTWARE

- Cisco Packet Tracer (any version)
- One Server
- Two PCs
- One Switch (2960)
- Straight-through Ethernet cables

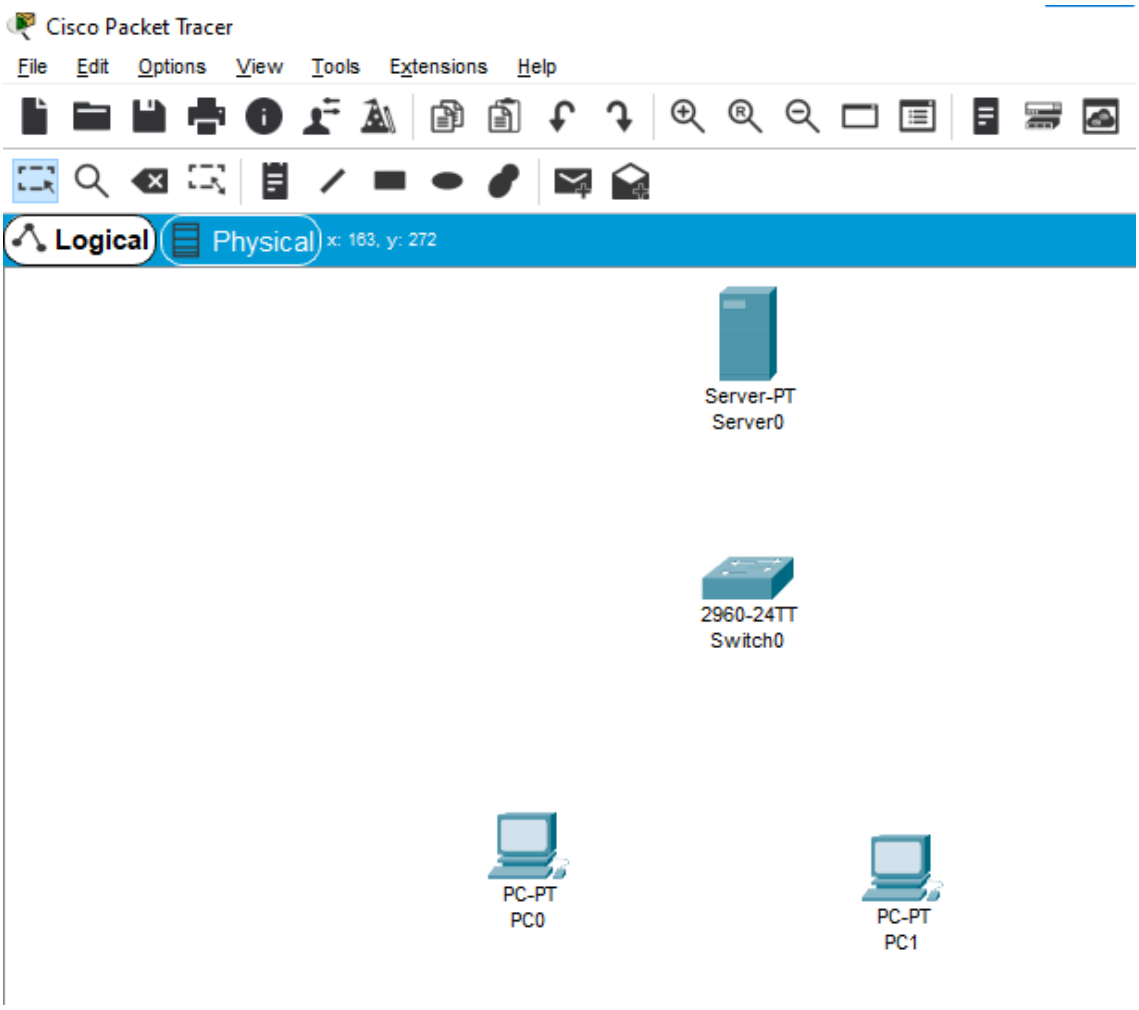
Step 1:- add server



Step 2:- add one switch

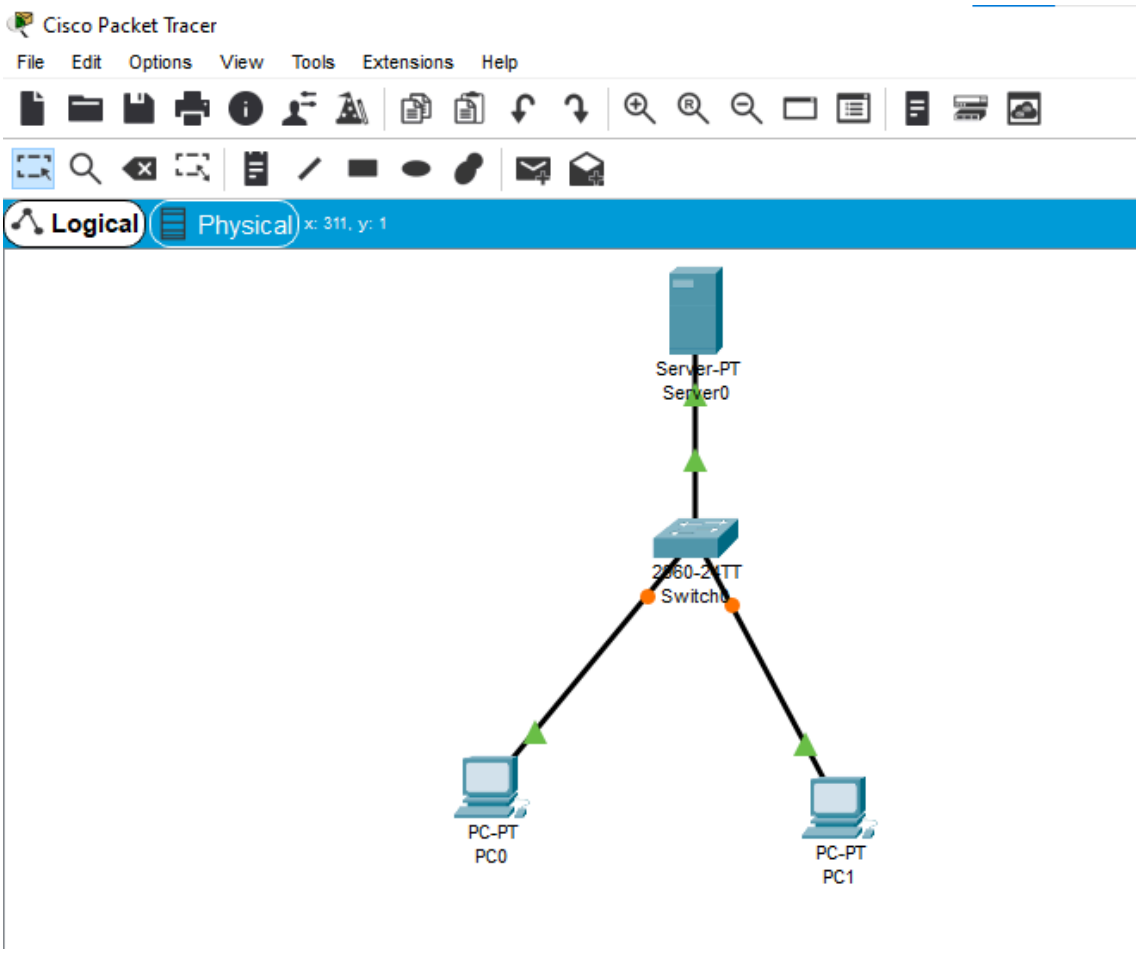


Step 3:- add 2 pc



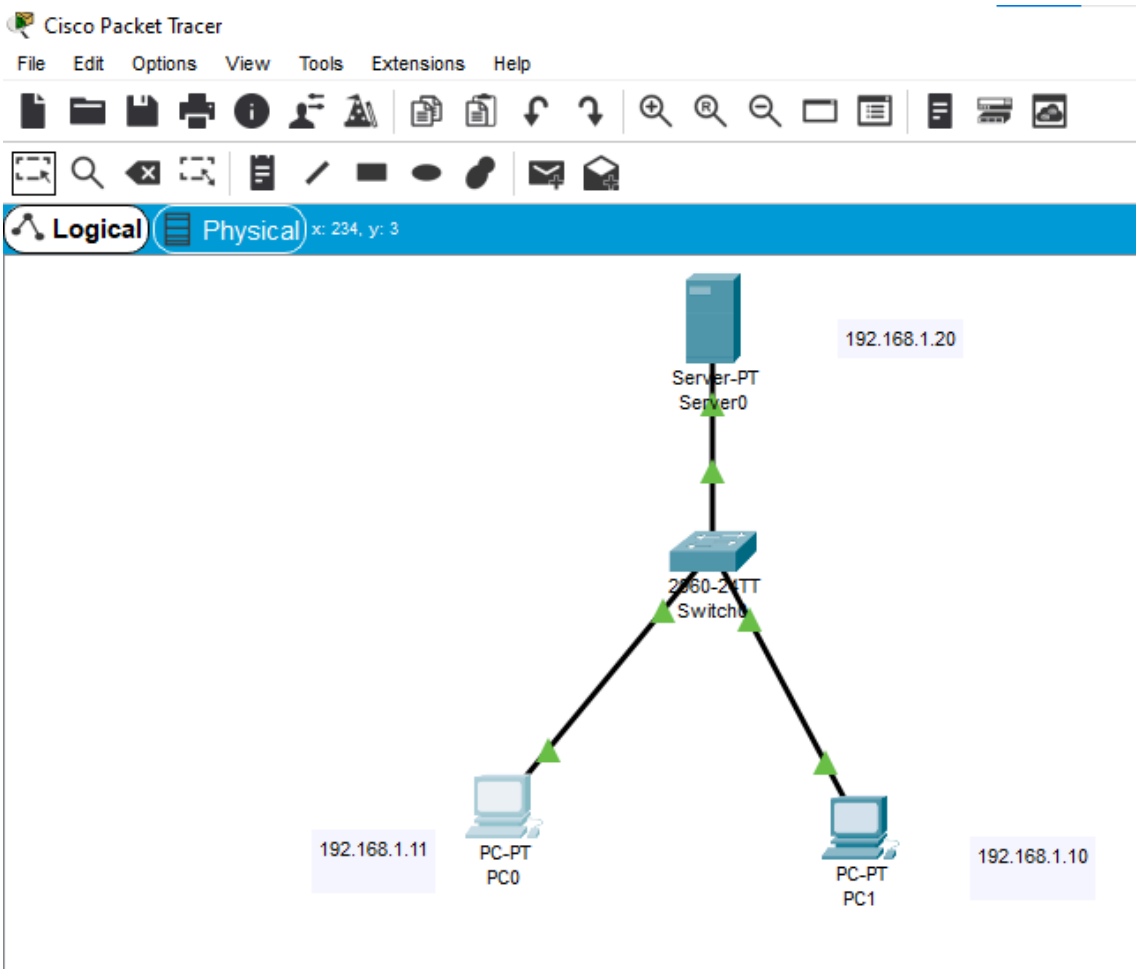
Step 4;-

Connect connection with wire



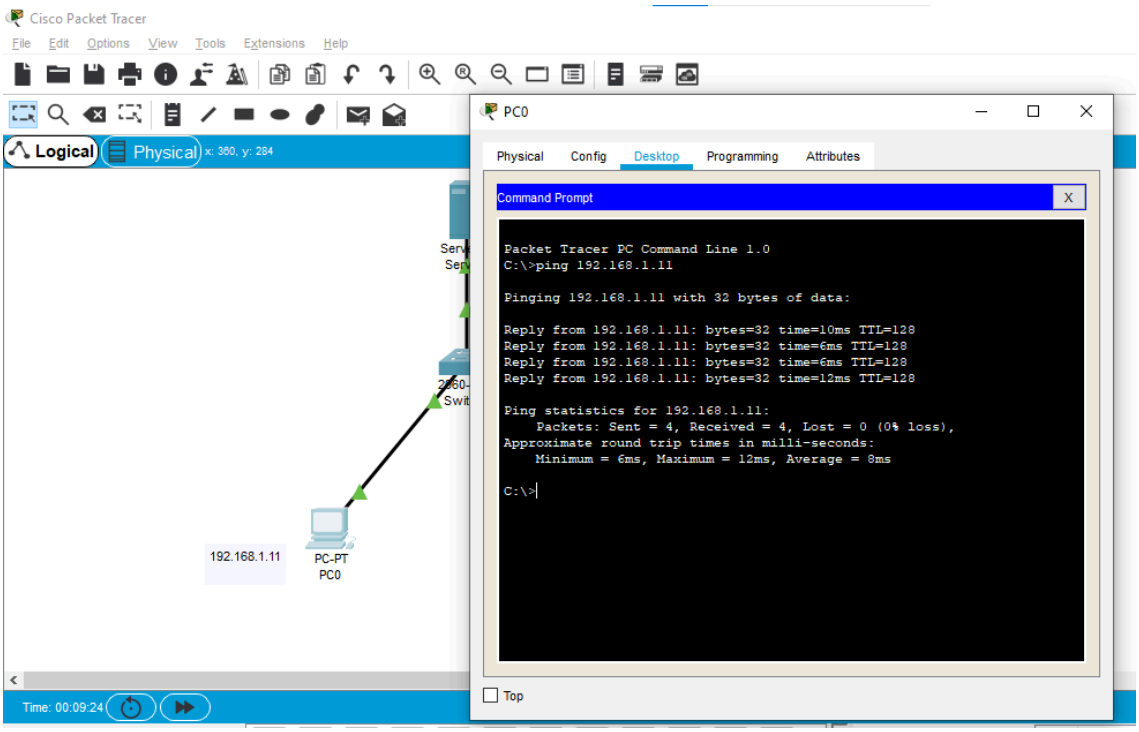
Step5:-

Give ip address to server and both the pc's



Step6:-

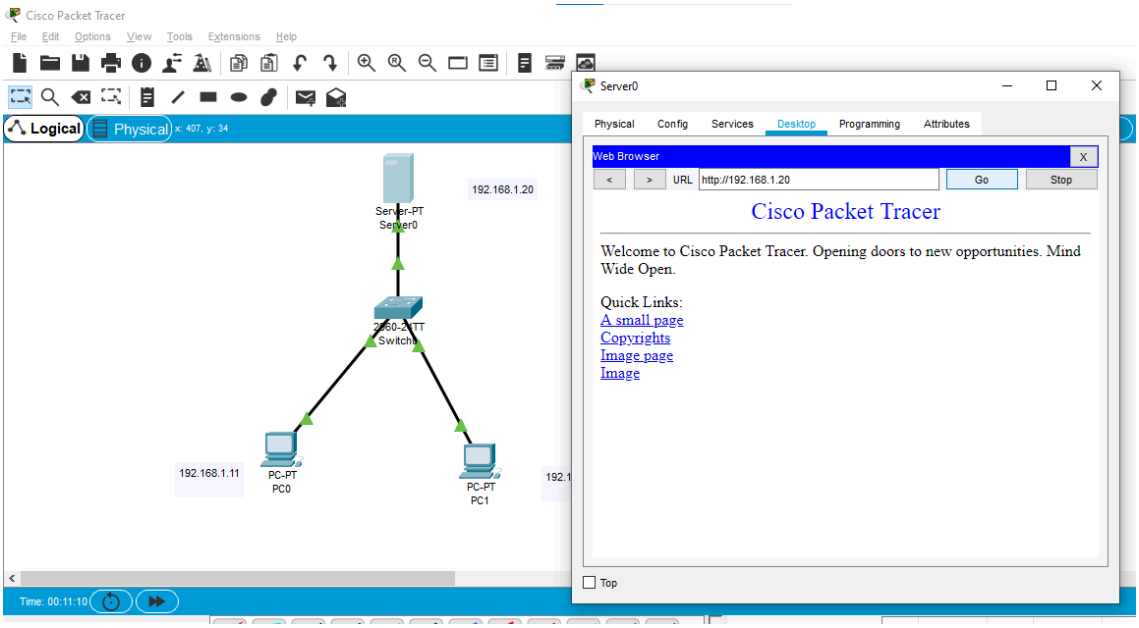
Check pc's are connected or not



Pc's are connected

Step 7:-

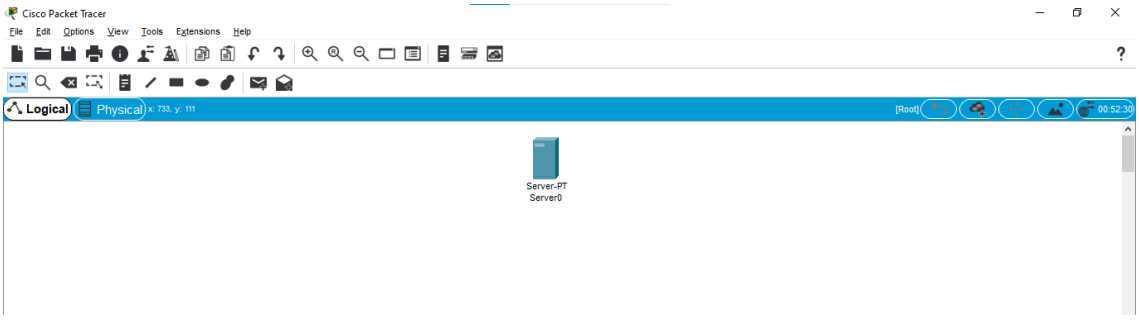
Check web page are connected or not



Practicalno. 04

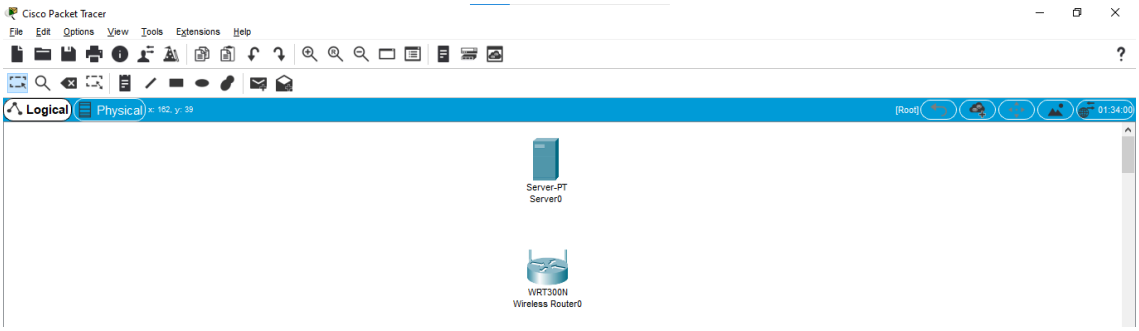
Step 01:-

Add server



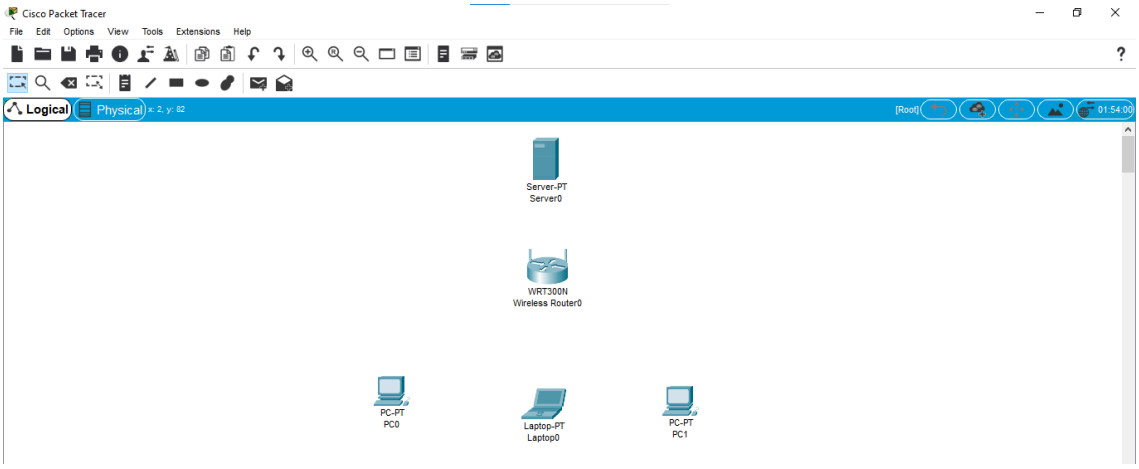
Step 02:-

add wireless router

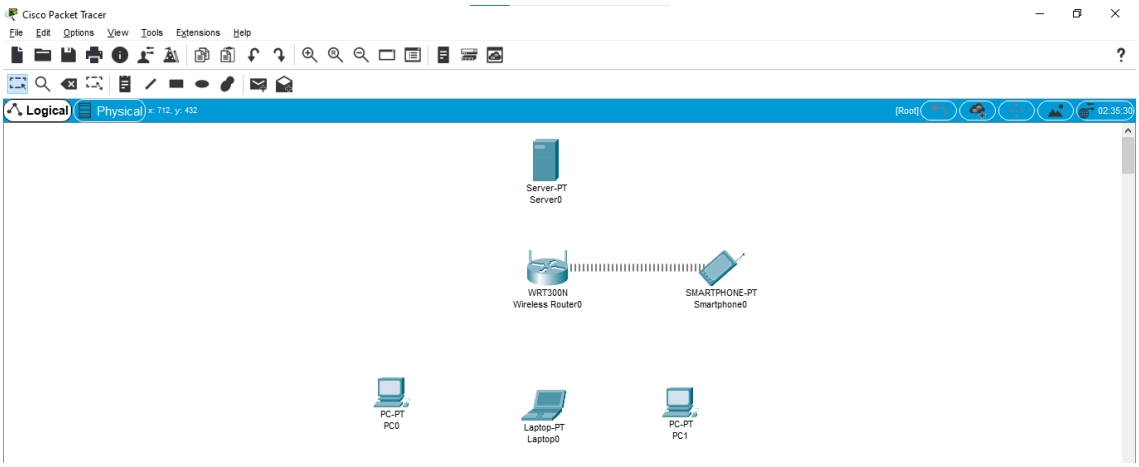


Step 03:-

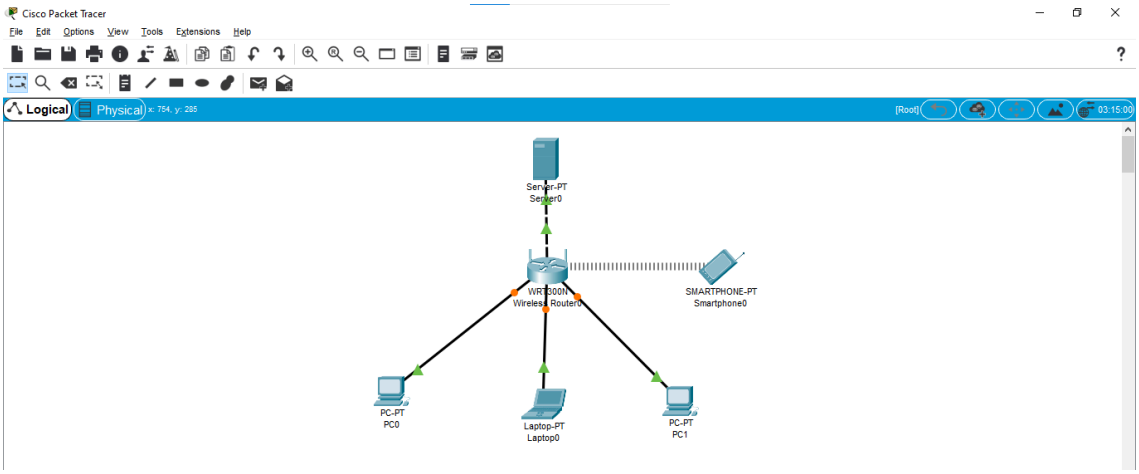
add two pc and one laptop



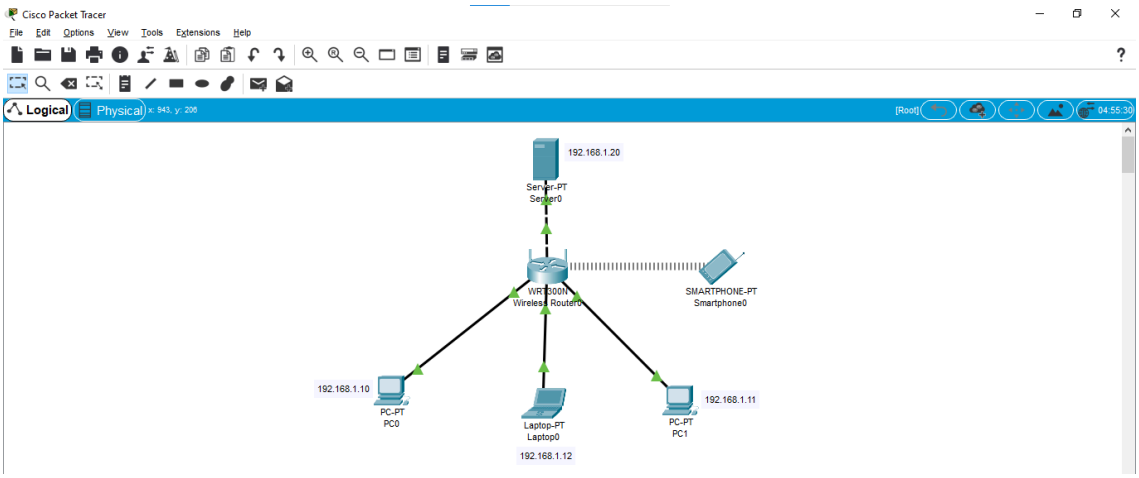
Step 04:-
add smart phone



Step 05:-
Then connect all smart devices through wire



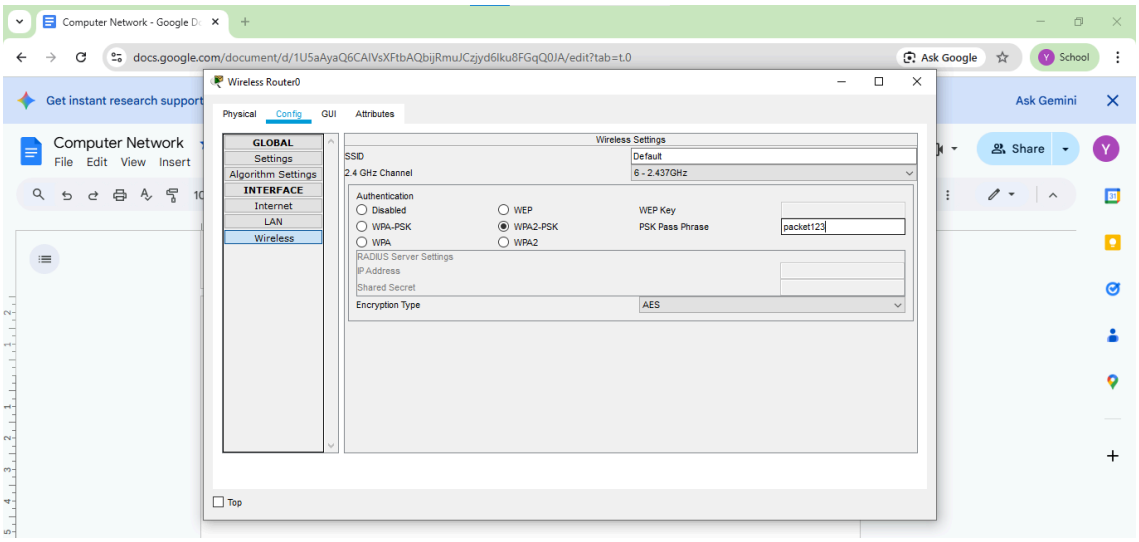
Step 06:-
Give ip address to all pc’s , server and laptop



Step 07:-

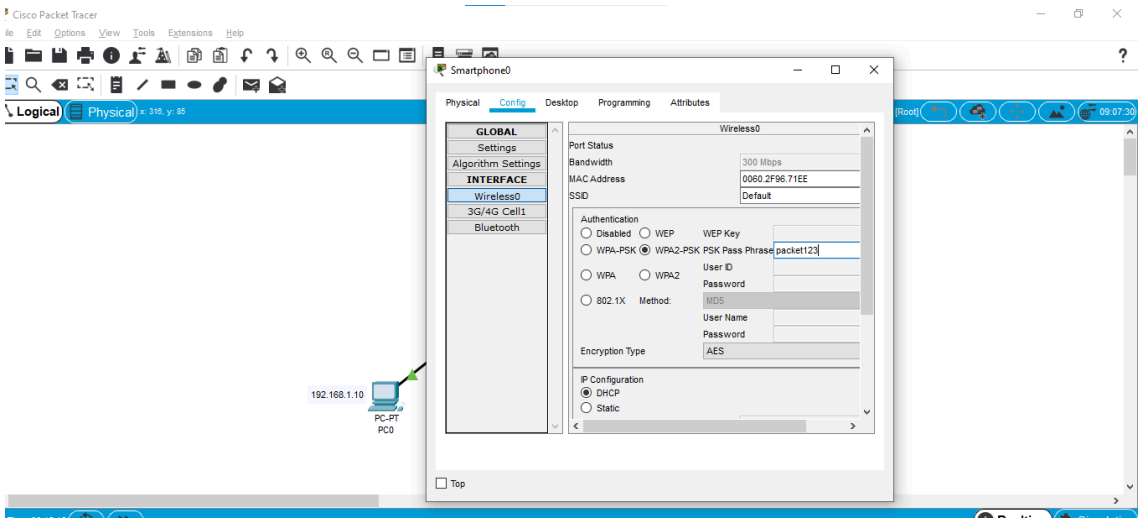
Now we have to check if the the wireless router and the smart phone are connected or not.

For that click on wireless router > wireless > then select option (WPA2-PSK) > then write “ packet123 “



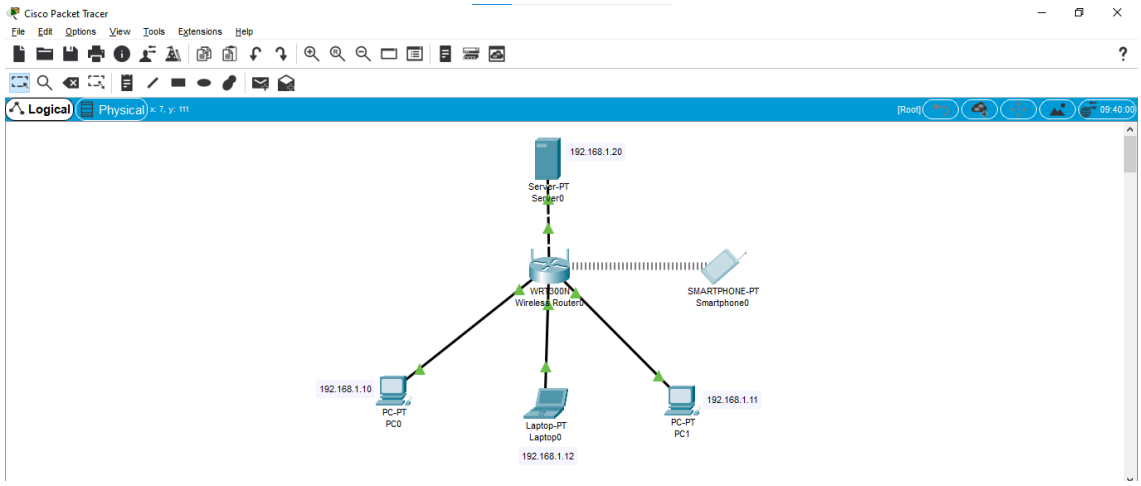
Step no.07:-

Then click on phone > config > wirelsee0 > then select option (WPA2-PSK) > then write “ packet123 “.



Step no08:-

Nowcheck if they are connected.

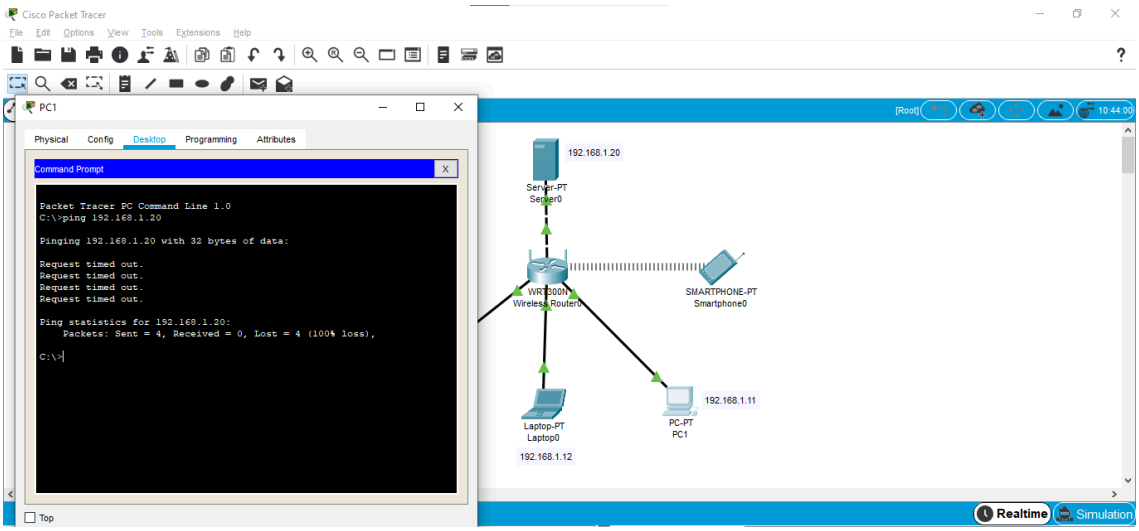


Connected

Step no 09:-

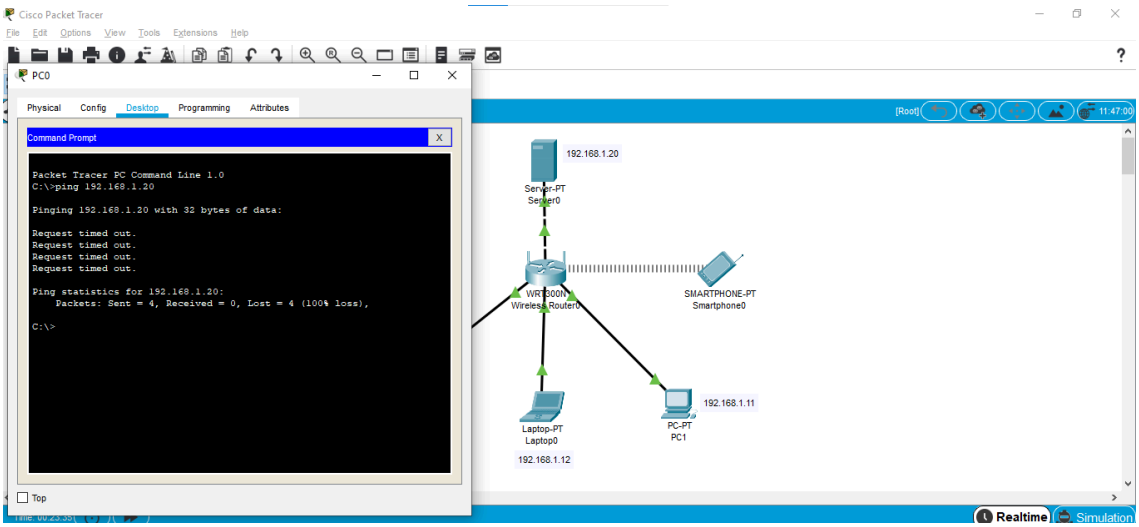
Now check the pc's and laptop are connected to the server or not.

1.For first pc



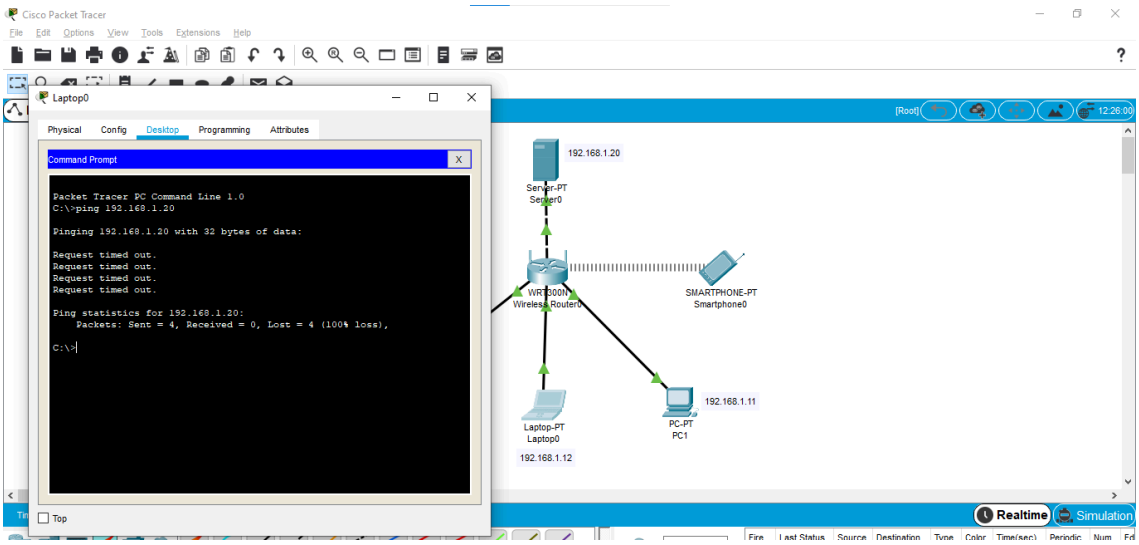
Connected

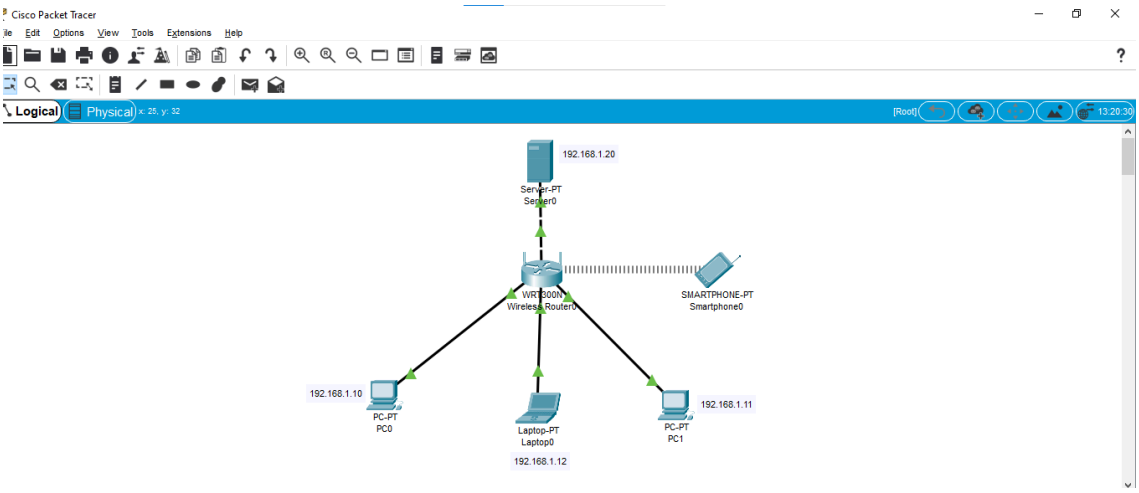
For 2 pc



connected

Now for laptop



Connected

All the thing are connected to the server